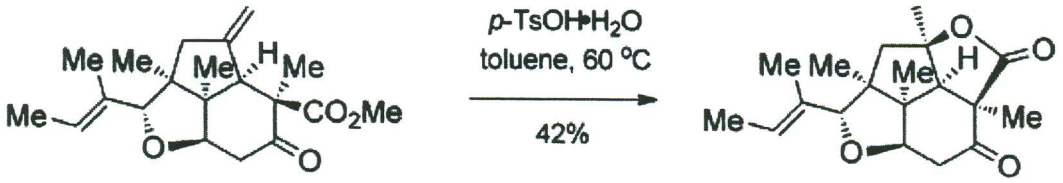
Organic Cumulative Exam November 14, 2014

All questions are based on the article:

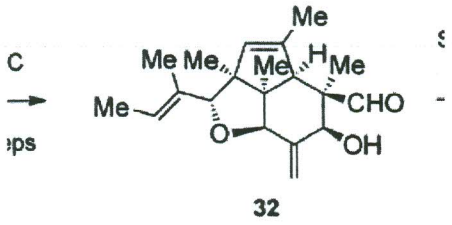
Stereoselective Total Synthesis and Structural Elucidation of (_-)-Indoxamycins A–F
Chi He, Chenlong Zhu, Bingnan Wang, and Hanfeng Ding
DOI: 10.1002/chem.201403986
Chem. Eur. J. **2014**, *20*, 15053 – 15060

1. Draw a step-by-step electron pushing mechanism for the conversion of **25** into **26**.

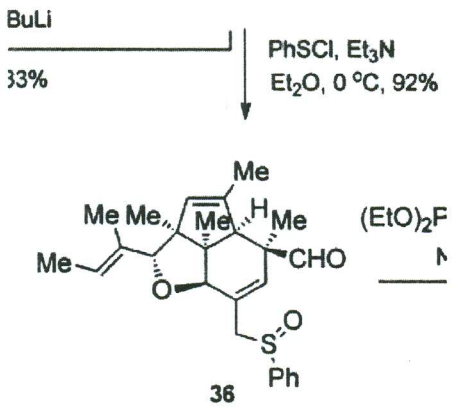


25 **26**

2. Draw a step-by-step electron pushing mechanism for the conversion of **32** into **36**.

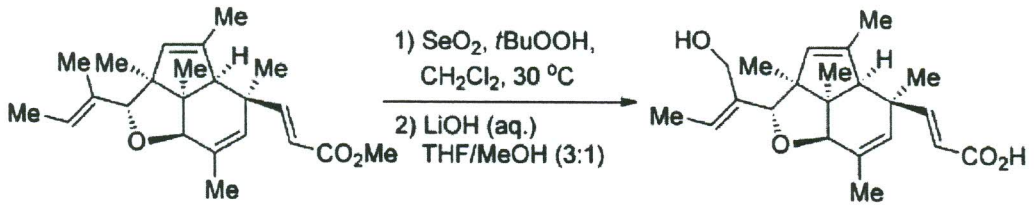


32



36

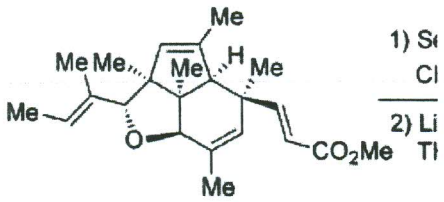
3. Draw a step-by-step electron pushing mechanism for the conversion of **38** into the methyl ester of (-)-indoxamycin D (i.e. ignore step 2).



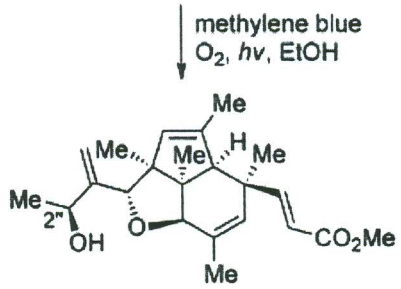
(-)-indoxamycin A methyl ester (**38**)

(-)-indoxamycin D ((1"*E*)-2-*epi*-**4**) [25%, revised structure]

4. Draw a step-by-step electron pushing mechanism for the conversion of **38** into **39**. Assume a reducing agent is added during the workup.



(-)-indoxamycin A methyl ester (**38**)



39 (45%) and 2"-*epi*-**39** (16%)